
RP-GAAP: Global Attribute-Association Pattern Aggregation with Rare-Pattern Weighting for Graph Fraud Detection

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Abstract

GAAP achieves state-of-the-art graph fraud detection by learning attribute-association patterns through adaptive feature binning and message passing. However, its standard cross-entropy loss treats every training node equally, causing rare fraud-indicative patterns to be drowned out by common benign ones. We propose RP-GAAP, which assigns higher sample weights to nodes with infrequent patterns, improving detection without any architectural modification. Preliminary results show Recall@K gains of +0.97 on T-Finance and +0.78 on Tolokers. Project page: <https://github.com/NathanTrance/GraphML>.

1 Introduction

Fraud on online platforms—fake reviews, money laundering, collusive crowdsourcing—causes billions in annual losses. Detecting these activities is inherently a graph problem: fraudsters interact with victims, form collusive rings, and leave topological traces invisible to non-relational models [11, 1]. Graph Neural Networks (GNNs) have become the dominant approach because they jointly model node attributes and neighborhood structure [3].

However, fraud detection presents two hard challenges for standard GNNs. First, camouflage: fraudsters mimic benign feature distributions to blend in [1]. Second, neighborhood inconsistency: a fraudster’s neighbors are mostly benign, diluting the fraud signal during message passing [6]. On top of these, class imbalance is severe—fraud typically constitutes 3–17% of nodes, so a model predicting “benign” for all achieves high accuracy while catching zero fraud. Existing detectors address these via neighbor filtering or sampling [1, 5, 6], adding complexity without treating pattern rarity itself. Focal loss [4] reweights by classification difficulty, not intrinsic rarity.

GAAP [2] departs from neighbor engineering by learning *attribute-association patterns*: adaptive feature binning preserves per-attribute semantics, message passing extracts neighborhood association context, and a global cross-attention module captures long-range correlations. GAAP achieves SOTA against 24 baselines on 7 datasets without neighbor filtering. However, it trains with standard cross-entropy, assigning equal weight to every node. Common benign patterns dominate the gradient, while rare fraud-indicative patterns receive negligible attention. Our central question: can we improve GAAP simply by telling the optimizer which training examples encode rare patterns?

We propose RP-GAAP (Rare-Pattern-weighted GAAP), a lightweight extension that computes per-sample weights from quantile-binned feature pattern frequencies and uses these scores in the cross-entropy loss. The method adds three hyperparameters, introduces no learned parameters, and requires no architectural changes. We evaluate RP-GAAP on four fraud detection benchmarks (YelpChi, T-Finance, Elliptic, Tolokers) across four loss configurations.

2 Proposed Method

2.1 Preliminaries

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an attributed graph with $N = |\mathcal{V}|$ nodes. Each node i has a feature vector $\mathbf{x}_i \in \mathbb{R}^d$ and a binary label $y_i \in \{0, 1\}$ (0 = benign, 1 = fraud). The nodes are partitioned into disjoint training, validation, and test sets. Fraud typically constitutes 3–17% of nodes, making the problem severely class-imbalanced.

2.2 Background: GAAP

GAAP [2] produces a node representation $\mathbf{h}_i \in \mathbb{R}^D$ for each node i through three stages, then predicts $\hat{y}_i = \text{softmax}(\mathbf{W}\mathbf{h}_i)$. It is trained with standard cross-entropy:

$$\mathcal{L}_{\text{GAAP}} = -\frac{1}{|\mathcal{T}_{\text{train}}|} \sum_{i \in \mathcal{T}_{\text{train}}} \log \hat{y}_{i, y_i} \quad (1)$$

2.2.1 Dynamic Binning Embedding (DyBEM)

Raw continuous features cannot be fed directly into message passing without fusing distinct semantics. DyBEM addresses this by learning b adaptive bin boundaries per feature dimension. For the j -th feature, the bin boundaries $\{q_{j,0}, q_{j,1}, \dots, q_{j,b}\}$ are initialized via decision-tree binning on the training set [2] and optimized jointly with the GNN. Each scalar value $x_{i,j}$ is then represented as a b -dimensional vector where the m -th component encodes the fragment-wise membership of $x_{i,j}$ in bin m , producing a bin embedding $\mathbf{e}_{i,j} \in \mathbb{R}^b$. Features from all d dimensions are stacked to form $\mathbf{E}_i \in \mathbb{R}^{d \times b}$, which is flattened and projected to the GNN’s hidden dimension.

2.2.2 Message-Passing with GraphSAGE

GAAP employs GraphSAGE [3] with max-pooling aggregation as its GNN backbone. For each node v at layer $l + 1$, the update rule is:

$$\mathbf{h}_v^{(l+1)} = \sigma \left(\mathbf{W}_{\text{self}}^{(l)} \mathbf{h}_v^{(l)} + \mathbf{W}_{\text{neigh}}^{(l)} \cdot \max_{u \in \mathcal{N}(v)} \sigma_{\text{pool}}(\mathbf{W}_{\text{pool}}^{(l)} \mathbf{h}_u^{(l)}) \right) \quad (2)$$

where $\mathbf{h}_v^{(0)} = \text{MLP}(\mathbf{E}_v)$ is the DyBEM-projected feature, $\mathcal{N}(v)$ denotes the neighborhood of v , and σ is a ReLU activation. Max-pooling is chosen because it preserves the strongest signals per dimension rather than averaging them away—critical when a fraud node is surrounded by mostly benign neighbors. The output of the final GNN layer is denoted $\mathbf{g}_i \in \mathbb{R}^{d_h}$, representing the node’s local *association pattern*.

2.2.3 Global Pattern Aggregation

A multi-head cross-attention module then captures long-range correlations by attending each node’s current GNN output against historical embeddings cached from the previous epoch [2], mitigating over-smoothing while keeping computation efficient.

2.3 RP-GAAP: Rare-Pattern Weighted Training

Under GAAP’s equal-weighted loss (Eq. 1), all training nodes contribute identically to the gradient. Yet the distribution of attribute-association patterns is highly skewed: common patterns dominate, while rare patterns—disproportionately associated with fraud—receive negligible gradient signal. RP-GAAP addresses this by precomputing a rarity weight w_i for each training node.

Pattern construction. We select the top- k features by variance over $\mathcal{T}_{\text{train}}$. For each selected feature j , we partition its training-set values into b quantile bins with boundaries $q_{j,0} < \dots < q_{j,b}$. Node i is assigned bin index $c_{i,j}$ where $q_{j,c_{i,j}} \leq x_{i,j} < q_{j,c_{i,j}+1}$. The k bin indices are combined into a single pattern identifier:

$$p_i = \sum_{j \in \mathcal{S}} c_{i,j} \cdot b^{\pi(j)} \quad (3)$$

where $\mathcal{S}$ is the set of selected feature indices and $\pi(j)$ maps each feature to its position in the set.

Weight assignment. Let $f(p) = |\{i \in \mathcal{T}_{\text{train}} : p_i = p\}|$ be the frequency of pattern p in the training set. We linearly interpolate between 1.0 (most frequent) and w_{max} (rarest), then apply a multiplicative boost $\phi \geq 1$ to fraud nodes, clipped to $[1.0, w_{\text{max}} \cdot \phi]$:

$$w_i = \min\left(\left[1.0 + (w_{\text{max}} - 1.0) \cdot \frac{\max_q f(q) - f(p_i)}{\max_q f(q) - \min_q f(q)}\right] \cdot \phi^{\mathbb{1}_{[y_i=1]}}, w_{\text{max}} \cdot \phi\right) \quad (4)$$

Nodes not in $\mathcal{T}_{\text{train}}$ receive $w_i = 1.0$. The weight vector $\mathbf{w} \in \mathbb{R}^N$ is computed once before training on CPU (adding <1 second overhead) and requires no gradients.

RP-GAAP loss. We replace the uniform average in Eq. 1 with a weighted sum:

$$\mathcal{L}_{\text{RP-GAAP}} = -\frac{1}{\sum_{i \in \mathcal{T}_{\text{train}}} w_i} \sum_{i \in \mathcal{T}_{\text{train}}} w_i \cdot \log \hat{y}_{i, y_i} \quad (5)$$

The GAAP forward pass (DyBEM $\rightarrow$ GraphSAGE $\rightarrow$ MHA $\rightarrow$ classifier) is entirely unchanged. The method adds exactly 4 hyperparameters— $b, k, w_{\text{max}}, \phi$ —and no learned parameters. We use $b = 5, k = 10, w_{\text{max}} = 3.0, \phi = 1.5$ across all experiments.

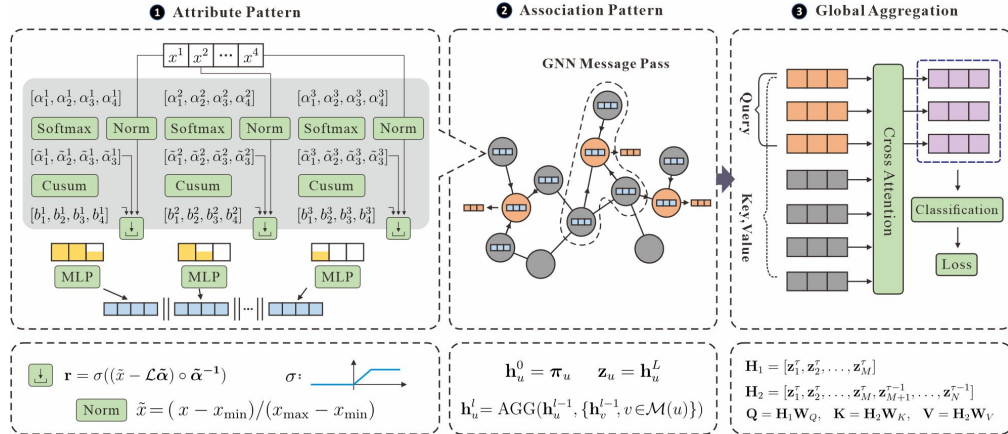


Figure 1: The GAAP framework. DyBEM creates binned attribute patterns, GraphSAGE (Eq. 2) extracts association patterns, and global cross-attention captures pattern correlations. RP-GAAP adds rarity weights (Eq. 4) to the loss without modifying this architecture. Figure adapted from [2].

3 Experimental Setup & Preliminary Results

3.1 Datasets and Metrics

We evaluate on four GADBench [10] benchmarks following GAAP’s splits [2]: **YelpChi** [8] (review spam, 14.5% fraud), **T-Finance** [9] (financial transactions, 3.4% fraud), **Elliptic** [12] (illicit Bitcoin, 9.5% fraud), and **Tolokers** [7] (crowdsourcing, 16.9% fraud). Following GAAP, our primary metric is Recall@K—the fraction of true fraud nodes in the top- K predictions where K equals the total number of fraud nodes; we also report AUC-ROC, AP, Macro-F1, and fraud-class Recall and Precision.

Table 1: Recall@K on four datasets. Best per dataset in bold.

Dataset	GAAP	+Focal	RP-GAAP	+Both	Best Δ
YelpChi	0.8839	0.8823	0.8862	0.8700	+0.23
T-Finance	0.8488	0.8530	0.8585	0.8571	+0.97
Elliptic	0.7322	0.7359	0.7313	0.7322	+0.37
Tolokers	0.5498	0.5296	0.5467	0.5576	+0.78

3.2 Preliminary Results

Table 1 reports Recall@K with default hyperparameters across all four loss configurations.

RP-GAAP improves over the GAAP baseline on YelpChi (+0.23) and T-Finance (+0.97), the dataset with the most extreme class imbalance. On Tolokers, combining rare weighting with focal loss yields the best Recall@K (+0.78). As next steps, we will run ablation studies and perform hyperparameter tuning to yield the final results.

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